Aswanth Bindu Ajith

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1PROFESSIONAL SUMMARY

M.Sc. student in High Performance Computing and Quantum Computing with a Mechanical Engineering foundation. Comfortable building high-performance computing, software engineering solutions from low-level numerical code to production web features. Reduced simulation runtime by 40

2TECHNICAL SKILLS

HPC Parallel Computing: MPI (basic), OpenMP (basic), GPU-aware computing concepts, Cluster job scheduling (SLURM basics), Performance profiling

Machine Learning: PyTorch, Scikit-learn, TensorFlow (basic), Feature engineering, Model evaluation

Programming Languages: Python (Advanced, 4+ years), C/C++, JavaScript, Bash/Shell scripting, HTML/CSS

Scientific Computing: NumPy, Pandas, SciPy, Matplotlib, Jupyter, LaTeX

3PROFESSIONAL EXPERIENCE

Working Student Software Developer — *Germany*
No Border (E-commerce) — Feb 2025 Present

- Shipped 5+ frontend features in a 2-week sprint cycle, reducing average ticket age by 30
- Debugged and resolved 20+ production bugs across the React/Vue stack, cutting reported error rates by 25
- Wrote unit tests for 3 core modules, pushing test coverage from 45
- Collaborated with 4-person cross-functional team (design, backend, QA) using GitHub Flow and daily standups

4EDUCATION

M.Sc. High Performance Computing and Quantum Computing — 2024 Present
Deggendorf Institute of Technology (DIT), Germany — Current Average: 2.6
Relevant coursework: Parallel Distributed Computing, Numerical Optimization, Quantum Algorithms, Scientific Software Engineering, High-Dimensional Data Analysis
Highlights: Optimization Methods (1.7); System Design Applications (1.7); Software Engineering (2.3); Advanced Mathematics (2.3)

B.Tech Mechanical Engineering — 2019 2023
APJ Abdul Kalam Technological University, India — CGPA: 6.48

5PROJECTS

Large-Scale Computational Simulation Data Analysis

- Built a Python simulation engine processing 30,000+ interacting entities using vectorized NumPy operations
- Reduced runtime from 12 minutes to 7 minutes (40
- Identified 3 non-obvious data correlations that shifted the team's modeling assumptions
- Packaged results into an 18-page LaTeX report with reproducible Jupyter notebooks for peer review
- Key metrics: 30,000+ entities, 40

Computing Architecture Optimization Methods

- Implemented gradient descent and genetic algorithms from scratch to optimize a 50-dimensional engineering cost function
- Benchmarked CPU vs. GPU (CUDA) performance on matrix operations, documenting 8x speedup for large matrices
- Profiled memory bottlenecks using Python's cProfile and reduced peak RAM usage by 35
- Presented findings to a 25-person seminar, receiving top marks on the technical report
- Key metrics: 50-dimensional optimization, 8x GPU speedup, 35

Thermal Analysis of Battery Modules

- Modeled heat transfer across a 12-cell lithium-ion battery pack using finite difference methods in Python
- Simulated 8 operating scenarios, identifying a thermal runaway risk at >45řC under sustained 3C discharge
- Proposed a revised cooling channel geometry that improved peak temperature uniformity by 18
- Co-authored a technical report cited by 2 subsequent student projects in the same lab
- Key metrics: 12-cell pack, 8 scenarios, 18

6RESEARCH INTERESTS

Scientific Machine Learning, Parallel and Distributed Systems, High Performance Computing, Quantum Computing

7LANGUAGES

English (Professional Working Proficiency), Malayalam (Native), German (A1 (Beginner))